



David Munoz Tord

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Skills

R	10 yrs
SQL	5 yrs
Python	5 yrs
Bash	7 yrs
JavaScript	5 yrs

Libraries

- ▶ Shiny, Quarto, R Markdown
- ▶ devtools, testthat, roxygen2, renv
- ▶ tidyverse, data.table, ggplot2, plotly, Highcharts
- ▶ brms, lmer, Rstan, JAGS, H2O, XGBoost, tidymodels
- ▶ Scikit-learn, PyTorch

Technologies

- ▶ Git - BASH
- ▶ Jira - Agile
- ▶ CI/CD - Docker
- ▶ Posit Server - AWS
- ▶ REST APIs - Postgres

Domains

- ▶ Clinical Trial Reporting
- ▶ CDISC (SDTM/ADaM)
- ▶ Data Mining
- ▶ Machine Learning
- ▶ Data Visualization
- ▶ Bayesian Statistics

Profile

Senior R Developer with expertise in clinical trial reporting, software architecture, and production deployment. Specialist in the Shiny ecosystem, R package development, and CI/CD. Track record of generating CDISC-compliant TFLs and automating regulatory workflows using Quarto. Strongly involved in open source and technical mentoring.

Experience

Senior R Developer

02/2025 - Present

Cytel

- R Team at Johnson & Johnson managing the strategic transition from SAS: steering, evangelization, and continuous training of clinical teams
- Intensive development of documented (roxygen2) and tested (testthat) R packages for clinical trial programming, specifically aligned with CDISC standards
- Implementation of automated reporting systems leveraging Quarto to efficiently generate TFLs (Tables, Figures, Listings) for regulatory submissions
- Design, performance testing, and maintenance of analytical infrastructures (Posit Server) hosting Shiny applications in a strictly regulated environment

Data Science Consultant

01/2024 - 02/2025

Freelance - Pharmaceutical & Tech

- Led end-to-end data projects with full autonomy, mapping user requirements to scalable architecture and final deployment
- Developed high-performance custom R Shiny containerized dashboards, integrating REST APIs and implementing rigorous unit testing
- Predictive modeling and automation of data pipelines utilizing tidyverse and advanced data visualization libraries (ggplot2, echarts4r, plotly)

Full Stack Data Scientist

03/2023 - 01/2024

FlowBank

- Led the data team in evolving the architecture and creating the bank's analytical platforms (Data Science, Data Engineering, DevOps)
- Operationalized production-ready R Shiny applications (as packages) with continuous integration (CI/CD) and automated testing
- Maintained machine learning pipelines and provided daily technical mentoring to internal cross-functional teams

R Developer

04/2022 - 03/2023

FlowBank

- Developed internal analytical tools and automated reporting processes for business teams
- Created secure, documented, and tested internal R packages to meet banking sector constraints
- Conducted exploratory analysis of financial data and laid the groundwork for the internal R infrastructure

Lecturer in Biostatistics

09/2020 - 01/2022

University of Geneva

- Taught statistics and led practical workshops in R and Python
- Created and maintained course materials and technical documentation for students

PhD Candidate

03/2020 - 01/2022

Campus Biotech - Novartis

- Developed large-scale data processing pipelines for clinical and neuroimaging datasets
- Applied advanced Bayesian statistical modeling and machine learning methods
- Collaborated internationally and maintained rigorous data governance standards

Languages

French	Native
English	Fluent
Spanish	Native
Italian	Basic
German	Basic

Education

Ph.D. Candidate in Computational Neuroscience

03/2020 - 01/2022

EPFL - Switzerland

Topic: Food-related sensory alteration effects of GLP-1 receptor agonist liraglutide

MSc in Computational Neuroscience

09/2018 - 03/2020

University of Geneva - Switzerland

Thesis: Denoising of high-resolution fMRI signals using tICA

Complementary Studies in Computer Science

09/2017 - 06/2018

UMass Amherst - USA

Capstone: Machine Learning for neuroimaging data

BSc in Cognitive Science

09/2014 - 06/2017

University of Geneva - Switzerland

Thesis: Early detection of ASD event-related potentials

Extracurricular

- Top 12.5% on [GitHub](#) (2026)
- Top 2% on [StackOverflow](#) (2024)
- Global Fellow scholarship (2017)
- Co-founder of [Rocket-Courrier](#), an eco-responsible bike messenger cooperative.

Resume Source

[Quarto code](#)

Open Source Projects and Community

See my [GitHub profile](#) for a more comprehensive list.

Author & Lead Developer

R, Python, Shiny

- **junco** (Johnson & Johnson): Created a clinical trial reporting package (TFL generation, advanced statistical analyses, production-ready RTF/DOCX exports).
- **circle (rOpenSci)**: Official maintainer of the CircleCI integration package for R.
- **echarts4r** and **fiRebase (Opifex)**: Maintainer. Developed advanced interactive visualizations (R/JS) and implemented secure backend infrastructures (authentication, databases).
- **Cross-functional tools**: Creator of **DbVieweR** (database management in Shiny) and **MathAnimator** (mathematical animation engine integrating R and Python).
- **Major contributions**: Contributed source code to reference ecosystem packages, notably **ggplot2**, **brms**, **AFNI**.

Packages

Co-President

Training, Evangelization, Data Science

- **Typr**: Co-creator of a modern typing system for R designed to improve code safety and long-term package maintenance.
- **Open Source Training**: Designed and regularly hosted technical workshops for the community, making all my [workshops](#) freely available.
- Regularly organized training sessions, hackathons, and hosted the R community via “**R-Lunches**” at the University of Geneva.

We Data

Publications

See my [publications page](#) for a complete list of my scientific work.

Investigating the effect of liraglutide on self-reported liking and neural responses to food stimuli. Coppin, G., Munoz Tord, D., Pool, Cereghetti, D., Golay, A., Sander, D., & Pataky, Z. *Journal of Obesity* (2023) - [Paper](#)

Differential contributions of ventral striatum subregions to the motivational and hedonic components of the affective processing of reward. Pool, E. R., Munoz Tord, D., Delplanque, S., Stussi, Y., Cereghetti, D., Vuilleumier, P., & Sander, D. *Journal of Neuroscience* (2022) - [Paper](#)

3D-printed pacifier-shaped mouthpiece for fMRI-compatible gustometers. Munoz Tord, D., Coppin, G., Pool, E. R., Mermoud, C., Pataky, Z., Sander, D., & Delplanque, S. *Eneuro* (2021) - [Paper](#)

Early spatial attention deployment towards aggressive voices. Burra, N., Kerzel, D., Munoz Tord, D., Grandjean, D., & Ceravolo, L. *Social Cognitive and Affective Neuroscience* (2019) - [Paper](#)

Professional References

[Gabriel Becker](#) ✉, Statistical Computing Consultant and R-core contributor

[Joseph Kovach](#) ✉, Head of Technological Solutions at Johnson & Johnson

[John Coene](#) ✉, Lead Developer at Opifex

[Pierre Saouter](#) ✉, Head Data Scientist at FlowBank